

(Abstract)

Quarto Pandoc
LaTeX . Quarto (Multiline) ,

(Keywords): Quarto, LaTeX , PDF , Pandoc,

(Introduction)

PDF , (Robustness) . Quarto

(Scope of Study)

: 1. $O(N^3)$. 2. $\lim_{x \rightarrow \infty} f(x)$. 3.
(Unicode) .

(Mathematical Foundations)

Quarto PDF .

(Inline Math)

$x,$ $\mathbf{W} \in \mathbb{R}^{d \times k},$ $\mathbf{b} \in \mathbb{R}^k$ $y = \text{softmax}(\mathbf{W}^T \mathbf{x} + \mathbf{b})$.

(Aligned Integrals & Summations)

$J(\theta)$:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

(Maxwell's Equations)

LaTeX aligned :

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \text{Rot } \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \text{Rot } \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\end{aligned}$$

(Complex Matrices)

2 (Matrix) :

$$= \begin{pmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \cdots & \rho_{1n}\sigma_1\sigma_n \\ \rho_{21}\sigma_2\sigma_1 & \sigma_2^2 & \cdots & \rho_{2n}\sigma_2\sigma_n \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1}\sigma_n\sigma_1 & \rho_{n2}\sigma_n\sigma_2 & \cdots & \sigma_n^2 \end{pmatrix}$$

(System Architecture)

Quarto .

i						
PC	Quarto CLI	quarto	PATH	PDF	UI	.

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sequenceDiagram
    participant Editor as CM6 WYSIWYG Editor
    participant Bridge as Extended API Bridge
    participant CLI as Quarto CLI Subprocess

    Editor->>Bridge: compile_quarto_document(Content, Format)
    Note over Bridge:
    Bridge->>CLI: subprocess.Popen("quarto render ...")
    CLI-->>Bridge: stdout/stderr ( )
    Bridge-->>Editor:
```

```

CLI-->>Bridge: exit_code = 0 ( )
Bridge-->>Editor: { status: 'success', output_path: 'document.pdf' }
Note over Editor: iframe

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(Experiments)

LaTeX

(Format)	(LaTeX %)		(sec)	
HTML	100.0%	(CSS3)	2.12	(Responsive)
Report				
PDF	100.0%	(Vector)	4.87	(A4)
Document				
EPUB	94.2%		5.30	()
eBook				

(Conclusion)

, Quarto Pandoc LaTeX (XeLaTeX) .
 (BibTeX) .